

Supplement to Repair What Matters

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Supplement: Gaussian finite-edit display bound

For one protected pixel, let U be the unchanged ordered Gaussian support and E the complete edited support. Effective per-splat alpha is assumed to be post-clamp/post-cutoff and to satisfy $\alpha_i \in [0, 1]$. Per-channel emitted color after spherical-harmonic evaluation and the background lie in the same interval of width C_{color} .

For edited support E , define

$$A(E) = 1 - \prod_{i \in E} (1 - \alpha_i).$$

This is the maximum fraction of final transmittance whose contribution can be replaced, attenuated, or revealed through the edited splats. Since any channel value differs by at most C_{color} ,

$$\|R(U \cup E) - R(U)\|_{\infty} \leq C_{\text{color}} A(E).$$

For before/after edited sets E_0, E_1 , insert the common unchanged state U and apply the triangle inequality:

$$\begin{aligned} \|R(U \cup E_0) - R(U \cup E_1)\|_{\infty} &\leq \|R(U \cup E_0) - R(U)\|_{\infty} \\ &\quad + \|R(U) - R(U \cup E_1)\|_{\infty} \\ &\leq C_{\text{color}} (A(E_0) + A(E_1)). \end{aligned}$$

The output channel itself cannot change by more than its admissible interval width, hence

$$\|R(U \cup E_0) - R(U \cup E_1)\|_{\infty} \leq C_{\text{color}} \min\{1, A(E_0) + A(E_1)\}.$$

Insertion and deletion are obtained by taking one edited set to be empty. Depth-order changes involving edited splats are included because the edited set must contain every Gaussian whose projected opacity, color, or order relation can change. Renderer behavior outside these bounded assumptions is not softened into the certificate; it becomes an exact dependency or triggers FULL fallback. The image-space certificate is the maximum of the pixel bound over the declared protected support.